

12.421/12.621 Lecture 11: Atmospheric Absorption and Emission, Part 2

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Topics covered

- Atmospheric energy states
 - Vibrational energy
 - Electronic energy
 - Synthesis of energy states
- Broadening mechanisms w/ emphasis on pressure broadening

1 Absorption and emission by gases (cont. from prev. lecture)

- Recall that this is a very high-level discussion of absorption and emission by atmospheric gases
 - Students interested in a deeper dive should look into 12.315/12.815
- Recall from prev. lecture:
Total energy $\mathcal{E}$ of isolated molecule consists of 3 energy states

$$\mathcal{E} = \mathcal{E}_e + \mathcal{E}_v + \mathcal{E}_r, \quad (1)$$

$\mathcal{E}_e \equiv$ electronic energy $\rightarrow$ electron energy levels in individual atoms,

$\mathcal{E}_v \equiv$ vibrational energy $\rightarrow$ displacement between 2 or more nuclei,

$\mathcal{E}_r \equiv$ rotational energy $\rightarrow$ rotation of molecule.

★ Key point: energy states are quantized

- Rotational energy covered last lecture:

$$\mathcal{E}_r = \frac{\omega_{\text{rot}} \hbar \sqrt{J(J+1)}}{2}, \quad \begin{array}{l} \hbar \equiv (\text{reduced}) \text{ Planck const,} \\ J \equiv \text{nonnegative integer "rotational quantum number".} \end{array} \quad (2)$$

1.1 Vibrational energy $\mathcal{E}_v$

- Distance between atomic nuclei is influenced by absorption and emission of EM radiation
- Basic process (Figure 1)
- Vibration frequencies are quantized $\rightarrow$ “vibration levels”
- When molecule absorbs EM radiation w/ frequency associated w/ higher vibration level, “vibration transition” occurs

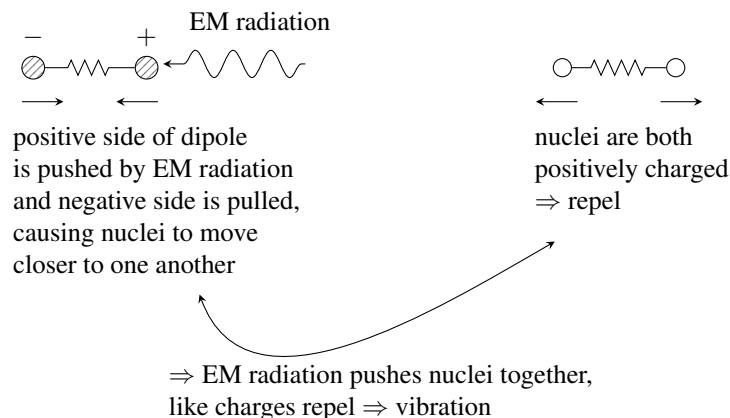


Figure 1: Basic process of molecular vibration driven by EM radiation.

- Vibration transitions involve higher energy levels than rotation transitions $\Rightarrow$ vibrational transitions are usually accompanied by rotation transitions
- Recall from last lecture that rotation transitions involve jumps of one and only one level (up or down) (Figure 2)

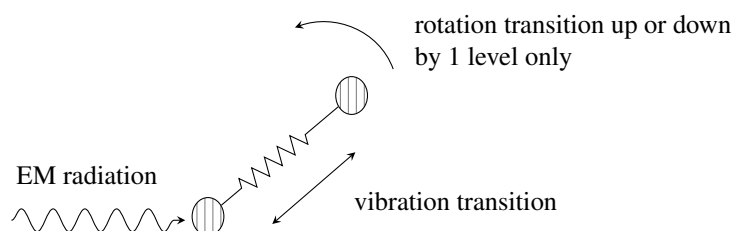


Figure 2: A vibration transition driven by EM radiation, accompanied by a rotation transition of one level.

- Since each vibration level can be associated w/ several rotation (sub-)levels, there are many possible vibration transitions within a population
- All vibration transitions require slightly different energy levels, which results in a series of closely spaced absorption lines that converge at short wavelength (i.e., higher freq) $\Rightarrow$ absorption bands in the intermediate infrared spectrum
- Details are well beyond scope of the course, and we will simply state that the energy levels for vibrations is

$$\mathcal{E}_v = \hbar\omega_{\text{vib},k} \left(n_{\text{vib},k} + \frac{1}{2} \right), \quad (3)$$

where $n_{\text{vib},k} \equiv$ (integer) vibrational quantum number for k normal mode
(normal mode: components of system moving sinusoidally w/ same freq and relative phase)

- Different types of molecules have different normal modes, sometimes called fundamentals

- To illustrate, consider composition of Earth’s atmosphere

$\text{N}_2 \sim 78.1\%$
 $\text{O}_2 \sim 20.9\%$
 $\text{Ar} \sim 0.9\%$
 $\text{CO}_2 \sim 0.04\%$
 other ($\text{H}_2, \text{Ne}, \text{He}, \text{CH}_4, \dots$)

Vibrational normal modes (Figure 3)

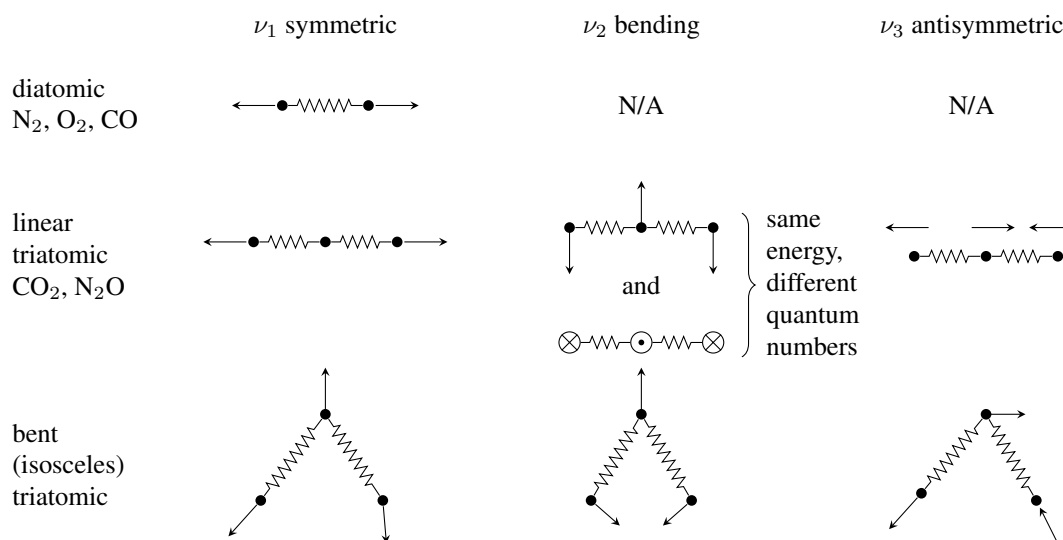


Figure 3: Vibrational normal modes of diatomic, linear triatomic, and bent triatomic molecules (after Liou 2002, Fig. 3.3). Crossed circles move into the page and the dotted circle out of it.

(see slides for movie of these for H_2O)

1.2 Electronic energy $\mathcal{E}_e$

- Represents electron energy levels within molecules
- Moving from one energy level to another requires absorbing or emitting a specific frequency of EM radiation

$$\Delta\mathcal{E} = \hbar\omega \quad (\text{Bohr's formula}) \quad (4)$$

- Bohr postulated the quantum theory based on the observed hydrogen spectrum and by building on Planck’s oscillator concept

↪ Bohr postulated that the angular momentum L of an electron could only take discrete values

$$L = n_{\text{el}}\hbar, \quad n_{\text{el}} \equiv \text{principal quantum number} \quad (5)$$

↪ From this selection rule, Bohr showed (for the hydrogen atom)

$$\mathcal{E}_e \sim \frac{1}{n_{\text{el}}^2}, \quad n_{\text{el}} = 1, 2, 3, \dots \quad (6)$$

where the constant of proportionality is a universal constant

$$\mathcal{E}_e = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n_{\text{el}}^2} = -\frac{R_H h c}{n_{\text{el}}^2}, \quad (7)$$

$m_e \equiv$ mass of electron,

$e \equiv$ charge of electron,

$h = 2\pi\hbar$,

$\epsilon_0 \equiv$ permittivity of free space,

$R_H \equiv$ Rydberg constant for hydrogen.

1.3 Putting it all together

(See Liou “An Introduction to Atmospheric Radiation” Ch. 1–3 for more details)

- Electronic and vibrational energy are closely related because both come from valence bonds that bind atoms into molecules
- Sign and magnitude of force between two atoms in a molecule depends on 2 factors:
 - distance between nuclei
 - electron energy level
- To illustrate, let us consider a simple potential energy curve for two electronic states and multiple possible vibration levels of a diatomic molecule (Figure 4)

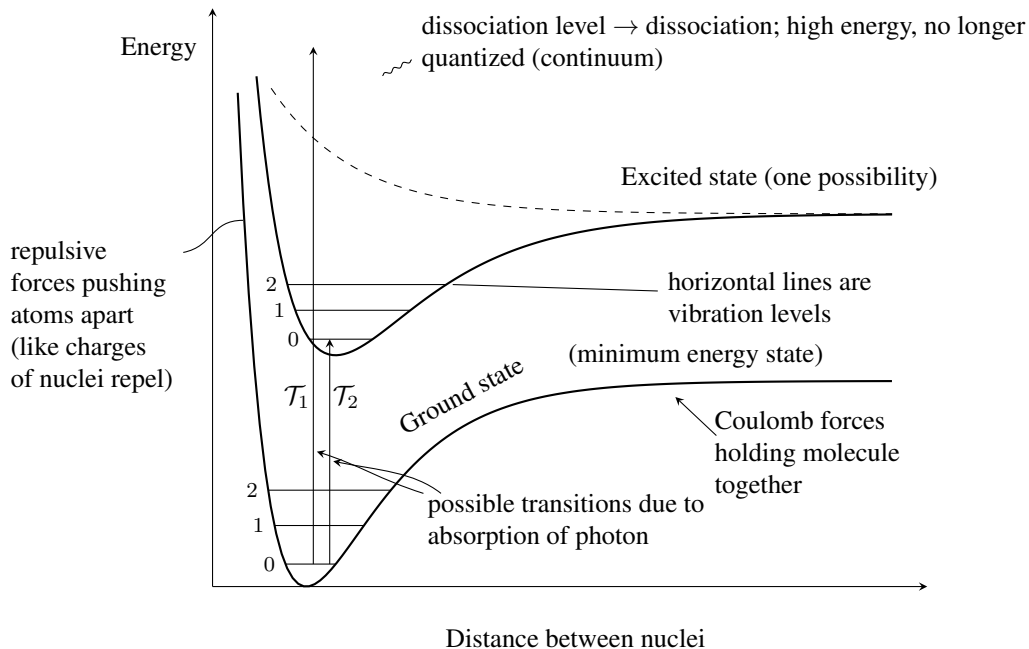


Figure 4: Illustrative potential energy curves for two electronic states of a diatomic molecule, with vibration levels (horizontal lines) and two possible transitions due to absorption of a photon (after Liou 2002, Fig. 3.4). Numbers label the vibrational quantum number n_{vib} .

- Key points from potential curve
 1. Energy gain from ground state to excited state (due to electronic transition) is simply the difference in minimum potential energies
 2. Vibrations are represented as horizontal lines showing changes in distances between nuclei
 - ↔ vibration transitions (always accompanied by rotation transitions) entail change in energy level ⇒ range of oscillation
 3. Illustrated transitions $\mathcal{T}_1$ and $\mathcal{T}_2$
 - $\mathcal{T}_1$ from ground state to dissociation, which is not quantized ⇒ a continuum of absorption wavelengths near energy associated w/ $\mathcal{T}_1$
 - $\mathcal{T}_2$ remains quantized ⇒ transition to specific vibration level ⇒ absorption spectrum will have discrete wavelengths in absence of disturbances
 4. At atmospheric temperatures, most molecules are in ground vibrational states ⇒ transitions $\mathcal{T}_1$ and $\mathcal{T}_2$ are representative of transitions in atmosphere
 5. Most absorption processes in atmosphere are a combination of electronic and vibrational transitions as depicted
 - ↔ These generally occur in UV, visible, and near IR frequencies

2 Effects of disturbances on absorption (and emission) spectrum

- Previous discussion applicable to isolated, undisturbed, stationary molecular system
- But molecules are in constant motion, interacting and colliding w/ one another and w/ other matter (e.g., dust particles)
 - ⇒ Energy levels during transitions change slightly due to external influences on molecules ⇒ radiation emitted during repeated transitions is polychromatic (multiple frequencies), causing absorption spectra lines of finite width
- Increased absorption width called “line broadening”
- Two main causes of line broadening
 1. Collision or pressure broadening, which is dominant in troposphere
 2. Doppler broadening – results from different thermal velocities of atoms and molecules (Doppler effect) → only important in upper atmosphere where pressures are very low
- Pressure broadening is, in general, the dominant process influencing absorption and emission in Earth’s atmosphere

2.1 Pressure broadening

- We can understand pressure broadening by going back to our discussion of permittivity in Lecture 1 where we modeled polarization of materials as a damped harmonic oscillator. Same model applies for pressure broadening.
- Therefore, dielectric constant is

$$\varepsilon(\omega) = 1 + \frac{\omega_p^2}{(\omega_0^2 - \omega^2) + j\nu\omega}, \quad (8)$$

where

$\omega_p \equiv$ plasma frequency, $\omega_p^2 = \frac{Ne^2}{\epsilon_0 m_e}$,
 $e \equiv$ electron charge,
 $N \equiv$ number of dipoles per unit volume,
 $m_e \equiv$ mass of electron,
 $\epsilon_0 \equiv$ permittivity of free space,
 $\omega_0 \equiv$ natural frequency,
 $\nu \equiv$ damping coef.

- Defining the power absorption coef as we did for volume scattering, we have

$$\begin{aligned}
 \kappa_a &= 2k_0 n'' = -2k_0 \text{Im}\{\sqrt{\epsilon}\} \\
 &= \frac{S}{\pi} \frac{\nu/2}{(\omega - \omega_0)^2 + (\nu/2)^2} \rightarrow \text{sometimes called the Lorentz profile}
 \end{aligned} \tag{9}$$

where $S = \frac{\pi \omega_p^2}{2c} \equiv$ spectral line strength (near resonance, for a dilute gas).

- Similarly, the real component of the index of refraction is

$$n' = 1 - \frac{\omega_p^2}{4\omega_0} \frac{\omega - \omega_0}{(\omega - \omega_0)^2 + (\nu/2)^2} \tag{10}$$

- Plotting κ_a and n' (Figure 5)

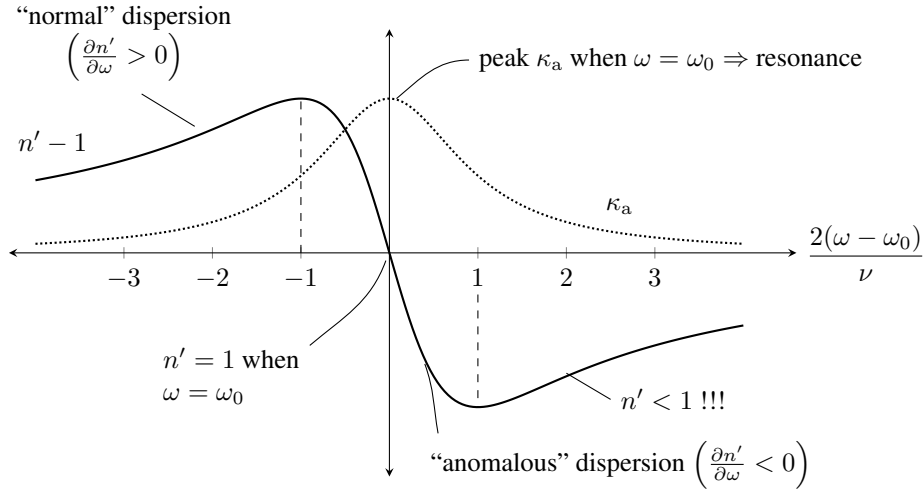


Figure 5: Absorption coefficient κ_a (dotted) and real part of the index of refraction, $n' - 1$ (solid), near a resonance at ω_0 (arbitrary units).

- Take-aways from κ_a and n' plot:

- For κ_a :

- 1) Peak absorption at resonance freq ω_0

2) width of κ_a defined by damping coef ν

→ what causes damping? emission (no apparent dissipative mechanisms)

But collisions play a role → collision of molecules causes phase shift in emitted EM wave. Collisions → random ⇒ population of molecules emit EM waves w/ random phases ⇒ reduced phase coherence ⇒ damping

– Since molecular collision plays a key role in damping, we may suppose a relationship between damping coef and temperature and pressure

$$\nu = \nu_0 \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right)^\delta \quad \text{for } \frac{1}{2} \leq \delta \leq 1 \quad (11)$$

$\nu_0 \equiv$ reference damping coef ($\nu = \nu_0$ for $P = P_0$ and $T = T_0$),

$P_0 \equiv$ atmospheric pressure at sea level,

$T_0 = 273$ K.

Plugging in temperature lapse rate for troposphere and P profile (see last lecture)

$$\frac{\nu}{\nu_0} = e^{-z/H_p} \left(1 - \frac{az}{T_0} \right)^{-\delta} \quad (\text{note sign}) \quad (12)$$

$H_p \simeq 8.4$ km,

$a \simeq 6.5$ K/km.

⇒ ν decreases to roughly $1/e$ of its surface value from surface to tropopause (~ 12 km altitude)

• For n' :

1. For most ω , $\frac{\partial n'}{\partial \omega} > 0$ → “normal” dispersion

(recall phase velocity $u_p = \frac{\omega}{k_0 n'} = \frac{c}{n'}$)

but near resonance freq ω_0 , $\frac{\partial n'}{\partial \omega} < 0$

→ “anomalous” dispersion → only occurs where κ_a is large and

$$\frac{\partial n'}{\partial \omega} = 0 \quad \text{when } \omega = \omega_0 \pm \frac{\nu}{2} \quad (13)$$

2. $n' = 1$ when $\omega = \omega_0$

3. Strikingly, $n' < 1$ when $\omega > \omega_0$ (in vicinity of ω_0 , not in general)

This implies $u_p > c$!!! called “superluminal” → many of you noticed this in question 1 of pset 1

Is it physical? Yes

Explanation:

- n' varies w/ ω ⇒ atmosphere is dispersive, at least in the neighborhood of ω_0
- Phase velocity only tells us how fast a wave of a given frequency travels (i.e., how fast a plane of constant phase travels)
- In a dispersive material, waves of different frequencies travel at different speeds ⇒ wave packet distorts (Figure 6)



Figure 6: A wave packet distorts as it propagates through a dispersive material.

- Useful then to define the rate of propagation of a plane of constant amplitude

$$\rightarrow \text{group velocity } u_g = \frac{\partial \omega}{\partial k} \quad (14)$$

where $k = \omega/u_p = k_0 n'$ is the wavenumber in the medium ($k_0 = \omega/c$ in free space)

- Energy (information) travels at the group velocity (where absorption is weak, i.e., away from line centers) $\Rightarrow$ no requirement that $u_p \leq c$ even though $u_p < c$ is by far the most common situation (most of you will never encounter a case where $u_p > c$ but this example is useful to highlight $u_g = \partial \omega / \partial k$, an extremely useful concept that some of you will encounter in practice)

2.2 Doppler broadening

- Pressure broadening is the dominant process in much of the atmosphere, but in the upper atmosphere (stratosphere and beyond) pressures are low $\rightarrow$ pressures decrease w/ altitude while temperature increases w/ altitude $\Rightarrow$ damping due to pressure broadening decreases rapidly w/ altitude $\Rightarrow$ with increasing altitude through the stratosphere and above, Doppler broadening becomes increasingly important and eventually dominant (for infrared lines, both matter from about 20 to 50 km; microwave lines remain pressure broadened to higher altitudes)
- From the kinetic theory of gases and assuming thermodynamic equilibrium, it can be shown that the absorption coef for Doppler broadening is

$$\kappa_{a,D} = \frac{S}{\alpha_D \sqrt{\pi}} \exp \left\{ - \left(\frac{k_0 - k_{\text{line}}}{2\pi \alpha_D} \right)^2 \right\} \quad (15)$$

where

$$\alpha_D = \frac{k_{\text{line}}}{2\pi} \sqrt{\frac{2k_B T}{m_m c^2}}, \quad (16)$$

$k_B \equiv$ Boltzmann's constant,

$m_m \equiv$ molecular mass.

Here the line strength S is the integral of the absorption coefficient over $k/2\pi$; in Eq. (9) it is the integral over ω , so the two differ by the factor $2\pi c$.

2.3 Comparing pressure and Doppler broadening

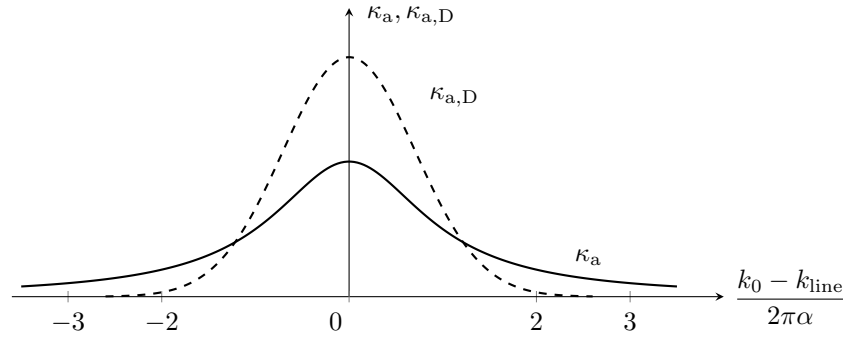


Figure 7: Pressure-broadened (Lorentz, κ_a , solid) and Doppler-broadened ($\kappa_{a,D}$, dashed) line shapes for the same line strength and width parameter (after Liou 2002, Fig. 1.11).

where

$$\alpha = \begin{cases} \frac{\nu}{4\pi c} & \text{pressure broadening,} \\ \alpha_D & \text{Doppler broadening.} \end{cases} \quad (17)$$

3 Brief summary

- Frequencies of absorption bands are determined by atmospheric composition and the mechanisms of molecular rotation, vibration, and electronic levels. These are in order of increasing energy $\Rightarrow$ vibration and electronic transitions are dominant and tend to manifest and cluster at higher frequencies (thermal IR + higher freq.) (at microwave frequencies, pure rotation lines of H_2O and O_2 dominate)
- Other mechanisms, namely collisions of molecules, broaden these absorption bands $\rightarrow$
- Together gives rise to the atmospheric transparency we've seen in class

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Constants</i>			
c	speed of light in a vacuum	m/s	2.998×10^8
h	Planck constant	J s	6.626×10^{-34}
$\hbar$	reduced Planck constant, $h/2\pi$	J s	1.055×10^{-34}
e	elementary charge (magnitude of the electron charge)	C	1.602×10^{-19}
m_e	electron mass	kg	9.109×10^{-31}
ε_0	permittivity of free space	F/m	8.854×10^{-12}
k_B	Boltzmann constant	J/K	1.381×10^{-23}
R_H	Rydberg constant for hydrogen	m^{-1}	1.097×10^7
<i>Molecular energy states</i>			
$\mathcal{E}$	total internal energy of a molecule	J	$\mathcal{E}_e + \mathcal{E}_v + \mathcal{E}_r$
$\mathcal{E}_e, \mathcal{E}_v, \mathcal{E}_r$	electronic, vibrational and rotational energy	J	–
$\Delta\mathcal{E}$	energy difference between two states (energy of the absorbed or emitted photon)	J	$\hbar\omega$
ω_{rot}	angular velocity of a molecule’s rotation	rad/s	–
J	rotational quantum number	dimensionless	0, 1, 2, ...
$n_{\text{vib},k}$	vibrational quantum number of normal mode k	dimensionless	0, 1, 2, ...
$\omega_{\text{vib},k}$	natural angular frequency of normal mode k	rad/s	–
k	(subscript) index of a normal mode	–	–
ν_1, ν_2, ν_3	labels of the symmetric, bending and antisymmetric vibrational modes	–	–
L	angular momentum of an electron (Bohr)	J s	$n_{\text{el}}\hbar$
n_{el}	principal quantum number	dimensionless	1, 2, 3, ...
$\mathcal{T}_1, \mathcal{T}_2$	illustrated transitions (Figure 4)	–	–
<i>Line shapes</i>			
ω	angular frequency of the EM radiation	rad/s	–
ω_0	natural (resonant) angular frequency; line center	rad/s	–
ε	complex dielectric constant (relative)	dimensionless	–
ω_p	plasma frequency, $\omega_p^2 = Ne^2/(\varepsilon_0 m_e)$	rad/s	–
N	number density of dipoles	m^{-3}	–
ν	damping coefficient	s^{-1}	–
ν_0	reference damping coefficient, at P_0 and $T_0 = 273 \text{ K}$	s^{-1}	–
δ	temperature exponent of ν	dimensionless	$\frac{1}{2}-1$
κ_a	power absorption coefficient (pressure broadening)	Np/m	–
$\kappa_{a,D}$	power absorption coefficient (Doppler broadening)	Np/m	–
S	spectral line strength	varies	$\pi\omega_p^2/(2c)$ (Eq. (9))
k_0	free-space wavenumber, ω/c	rad/m	–
k_{line}	wavenumber at the line center, ω_0/c	rad/m	–

Symbol	Description	Units	Value
k	wavenumber in the medium, $\omega/u_p = k_0 n'$	rad/m	k_0 in free space
n', n''	real part and negative imaginary part of the index of refraction	dimensionless	–
u_p	phase velocity, $\omega/(k_0 n')$	m/s	c/n'
u_g	group velocity, $\partial\omega/\partial k$	m/s	–
α_D	Doppler width	m^{-1}	–
α	width parameter of a line (Eq. (17))	m^{-1}	–
m_m	molecular mass	kg	–
<i>Atmosphere</i>			
z	altitude	m	–
T	temperature	K	–
T_0	reference temperature of ν_0 (Eq. (11)); in $(1 - az/T_0)$, the sea-level standard temperature (Lecture 10)	K	273 (Eq. (11)); 288.15 (Lecture 10)
P	pressure	Pa	–
P_0	sea-level pressure	Pa	1.013×10^5
H_p	pressure scale height	m	≈ 8.4 km
a	temperature lapse rate	K/m	6.5 K/km